

DAY 3 | CONTENT DELIVERY PLAN

Network Security & Threats

Detailed Content & Interactive Delivery Plan — visual/table format, matching Day 2

■ Audience	■ Duration	■ Delivery style	■ Format
Complete beginners — same cohort as Day 2	9:00 AM–3:00 PM (6 hrs), 2 breaks + lunch + 3 labs	Visual storytelling, individual checkpoints, live Wireshark demo	reveal.js — fragments, before/after, evidence board

Workshop Design Overview



Day 3 = follow one packet on a journey — from a device, across the network, past the security controls watching it. Beginner depth, but by day's end everyone can open Wireshark and spot concrete red flags.

Core Learning Goals

1	OSI / TCP-IP Reason about security-relevant layers, not memorize all seven.
2	Packets, IPs & Ports Understand “who,” “where,” and “which service” in any conversation.
3	TCP vs UDP Know the practical difference and why it matters for security.
4	Core Protocols HTTP/HTTPS, DNS, SSH, SMB, RDP — ports and common abuse.
5	IoT Devices See always-on devices as an extension of the attack surface.
6	Firewalls / IDS / IPS Detect vs. prevent — know which control does what.
7	Wireshark Basics Read a PCAP: IPs, ports, protocols, traffic patterns.
8	Port Scan & C2 Beacon Recognize two concrete attack signatures in traffic.
9	Day 2 → Day 3 Link Trace the phishing case into today's network evidence.

How reveal.js Supports the Interaction



Fragments
Reveal one packet field at a time before explaining it.

✦	Before / After Animate a firewall, IDS, and IPS onto a flat network diagram.
✦	Branching choices “Which control stops this?” — each choice shows a different outcome.
✦	Embedded demo Live Wireshark window or recording during the capture section.
✦	Evidence board Build a PCAP investigation clue by clue.
✦	Vertical slides Optional depth (e.g., TCP handshake) below the main slide.



Design rule (Day 3 decision): every section ends with one individual checkpoint/exit-ticket question. All labs and exercises are solo, not group-based.

9:00 AM – 3:00 PM Flow

Time	Section	Delivery idea
09:00–09:15	Opening hook	“Follow the Packet” mystery journey + individual guess
09:15–09:55	Network Fundamentals	OSI/TCP-IP, packets, IPs, ports — “every packet tells a story”
09:55–10:05	Break	10 minutes
10:05–10:35	TCP vs UDP	Bank Transaction (TCP) vs. Video Call (UDP)
10:35–11:15	Common Protocols	HTTP/HTTPS, DNS, SSH, SMB, RDP
11:15–11:35	IoT & Connected Devices	Always-on devices as extra attack surface
11:35–12:20	Lunch	45 minutes
12:20–12:50	Firewalls, IDS vs IPS	Before/after network animation
12:50–13:20	Wireshark Intro	Live demo + guided walkthrough
13:20–13:50	Lab 1 – Wireshark Fundamentals	Individual PCAP exploration
13:50–14:20	Lab 2 – Threat Investigation	Individual: find Port Scan + C2 Beacon
14:20–14:35	Lab 3 – HTML Packet Challenge	Individual, no-Wireshark fallback format
14:35–14:45	Final Investigation – Friday Request Pt.2	Links Day 2's case to today's traffic
14:45–14:55	Closing Challenge	“Which packet looks suspicious?” — no safety net
14:55–15:00	Wrap-up	5-question recap + takeaway



1 Opening Hook: “Follow the Packet”

DURATION
15 min

INTERACTION
High — Individual

MAIN OUTCOME
Feel that every click sends something traceable across the network.

- 1 Welcome + goal: “Today we learn how information travels — and how attackers hide inside it.”
- 2 Show an animated packet leaving a laptop. Ask individually: where does it go, what does it carry?
- 3 Reveal fields one at a time: Source IP → Destination IP → Port → Protocol.
- 4 Closing line on screen: “Every packet tells a story.”



Facilitator message: not a full OSI lecture — just get everyone comfortable reading four fields (source, destination, port, protocol) before Lab 1.



2 Network Fundamentals: Packets, IPs & Ports

DURATION
40 min

INTERACTION
Medium

MAIN OUTCOME
Explain in plain language what an IP address and a port are.

Key Terms



OSI / TCP-IP (security view)

Only the layers that matter here: addressing (IP), transport (TCP/UDP + ports), application (HTTP/DNS, etc.).



Packet

A small data unit: header (addressing) + payload (content).



IP address

The “home address” of a device on the network.



Port

The “door number” at that address — identifies which service receives the traffic.



Protocol

The agreed rules two devices communicate under (HTTP, DNS, SSH...).

Slide Design (kept from your original)



Packet shown as a message traveling between two devices.



Fields appear gradually: Source IP → Destination IP → Port → Protocol.



Ends with: “Every packet tells a story.”



Individual checkpoint: label the IP vs. the port on a simple diagram.



3 TCP vs UDP

DURATION
30 min

INTERACTION
Medium-High —
Individual

MAIN OUTCOME
Explain, with a real example, why some traffic needs reliability and some needs speed.



TCP

- Connection-oriented, reliable, ordered
- Waits for confirmation (handshake)
- Used where accuracy matters: banking, web, email, files



UDP

- Connectionless, fast, no guarantee
- Sends without waiting
- Used where speed matters: video/voice calls, streaming, DNS



Security angle: UDP's lack of a handshake makes source-spoofing easier — basis for DNS amplification and similar flood attacks.



Individual checkpoint: for online banking, a live video call, and a software update — guess TCP or UDP and why, before the reveal.



4 Common Network Protocols

DURATION
40 min

INTERACTION
High — Individual

MAIN OUTCOME
Name the default port and the most common attack for each core protocol.

Slide Design (kept from your original)



Company network diagram with several devices, each showing its protocol.



Reveal a device: Protocol → Port → What it's used for.



Warning icon appears automatically when a device is shown using the protocol insecurely (e.g., RDP exposed to the internet).



Keep each protocol to three quick questions: What does it do? Where do we see it? Why does an attacker like it? Skip deeper detail — five protocols is already a lot for a beginner in one sitting.

Protocol	Port	Why an attacker likes it
HTTP / HTTPS	80 / 443	Credential interception on HTTP; lookalike HTTPS sites; SSL-stripping
DNS	53	Spoofing/cache poisoning; DNS tunneling; DNS amplification (DDoS)
SSH	22	Brute-force login; credential stuffing; weak/outdated keys
SMB	445	Unauthorized file-share access; lateral movement; unpatched exploits
RDP	3389	Brute-force login; direct internet exposure; ransomware entry point



5 IoT & Connected Devices (New Addition)

DURATION
20 min

INTERACTION
Light — Individual

MAIN OUTCOME

Name IoT examples and explain in one line why they widen the attack surface.

◆ Kept intentionally short — a bridge between “Protocols” and “Network Controls,” not a deep IoT module.

What Counts as IoT



Examples

Smart cameras, speakers, TVs, printers, smart locks, thermostats, wearables, sensors.

Why It Matters



Always online, rarely monitored like a laptop.



Default credentials

Many ship with a factory username/password (admin/admin) that owners never change — the single most common reason IoT devices get taken over.



Frequently left unpatched.



Each device = one more “door” (IP + open ports) to try.



A compromised device can become a quiet pivot point or botnet member.



Concrete example for the room: attackers scan the internet for exposed smart cameras still running factory-default logins, log straight in with the manufacturer's default password, and watch or hijack the feed — no hacking skill needed, just an unchanged password. This is the fastest way for a beginner to grasp “why attack IoT at all.”



Individual checkpoint: “Which of these devices would you be most cautious exposing directly to the internet, and why?” Takeaway on screen: “More connected devices = more doors to check.”



6 Network Security Controls: Firewall, IDS, IPS

DURATION
30 min

INTERACTION
Medium-High —
Individual

MAIN OUTCOME
Say, for a given attack, whether a firewall, IDS, or IPS
is the relevant control.



Confirmed: this before/after design matches the Day 2 phishing-markup pattern well — same staged-reveal logic, applied to network traffic instead of an email. The new IoT section above now gives it a natural “why so many doors need a gatekeeper” lead-in.

Staged Reveal

1

Before

All devices connected, no controls in place.

2

After: Firewall

Controls what's allowed in/out based on rules.

3

IDS

Detects and alerts on suspicious traffic that got through — detection only.

4

IPS

Detects and actively blocks it in real time — detection + prevention.



Memory hook: “IDS tells you, IPS stops it.”



Individual checkpoint: for each attack scenario shown, decide firewall / IDS / IPS before the reveal.



7 Packet Capture & Traffic Analysis

DURATION
30 min

INTERACTION
Live Demo — Easy

MAIN OUTCOME
Recognize the Wireshark interface and know what a
filter does before Lab 1.



Packet capture

What “capturing traffic” means and when it's used.



Wireshark interface

Packet list, packet details, packet bytes, filter bar.



Basic analysis

	Reading source/destination, protocol column, info column.
•	Filters Filter by IP, port, or protocol to investigate specific traffic.



“Don't panic.” Say this before opening Wireshark. Nobody is expected to understand every row. Only four things matter today: Source, Destination, Protocol, Port — everything else is noise for now.

The “Wow Moment”

✨	Open Wireshark live, start a capture, then open a normal website in front of the room.
✨	Packets start flooding in instantly — let the room react before explaining anything.
✨	Land the point: “Every single one of these happened because you opened one page. Now imagine a whole network.”



Then switch to the safe, pre-recorded sample capture for the guided walkthrough of one full packet, tying it back to Section 2's four fields. End exactly where Lab 1 begins.



8 Hands-on Labs (Individual)

All three labs are solo, per the Day 3 decision. Each ends with a short checkpoint before moving on.

Lab 1 — Wireshark Fundamentals

DURATION	INTERACTION	MAIN OUTCOME
30 min	Hands-on — Individual	Independently open a PCAP and identify basic traffic details.

✓	Explore the Wireshark interface on their own machine.
✓	Open and inspect a provided PCAP file.
✓	Identify IPs, ports, protocols, and basic traffic patterns.
✓	Checkpoint Write down one source IP, one destination IP, and one protocol found.

Lab 2 — Threat Investigation

DURATION	INTERACTION	MAIN OUTCOME
30 min	Hands-on — Individual	Distinguish normal traffic from two specific attack patterns.

✓	Port Scan
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	Many connection attempts from one source to many ports/hosts in a short time.
✓	C2 Beacon Regular, repeating connections from an infected host to the same external address.
✓	Checkpoint Name the pattern found and the one indicator that convinced them.

Lab 3 — HTML Packet Inspection Challenge

DURATION 20 min	INTERACTION Hands-on — Individual	MAIN OUTCOME Apply the day's concepts without needing Wireshark itself.
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✓	Identify protocols from simplified packet data.
✓	Detect suspicious behavior.
✓	Differentiate legitimate vs. malicious traffic.
✓	Answer investigation questions per scenario.

◆	<i>Doubles as the no-setup fallback if Wireshark/PCAPs are unavailable — same role the Scavenger Hunt played on Day 2.</i>
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Companion Practice

	TryHackMe – Network Fundamentals Free room completed after/alongside Lab 1 to reinforce the day's concepts.
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9 Final Investigation: “The Friday Request — Part 2”

DURATION 15 min	INTERACTION High — Individual	MAIN OUTCOME Connect Day 2's social-engineering case to today's network evidence.
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Same Friday invoice case from Day 2 continues. The employee reported the message; the security team pulls the network logs.

Stage	Question	Debrief
1 – DNS lookup	What does this tell us?	Device tried to resolve the fake finance domain — confirms the link was touched.
2 – Outbound connection	What next?	Check destination IP, port, and protocol used.
3 – Repeating connections	What pattern is this?	A C2 beacon — possible check-in with an attacker server.

4 – Firewall/IDS logs	Who saw it, who could've stopped it?	IDS alerted; a proper firewall/IPS rule could have blocked it.
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9b Closing Challenge: Spot the Suspicious Packet

DURATION 10 min	INTERACTION High — Individual	MAIN OUTCOME Apply the day's reading skills one last time before the recap, with no safety net.
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1	Show a single screenshot: a PCAP snapshot with 10–15 packets, no explanation given.
2	Ask each participant individually: “Which packet looks suspicious — and why?”
3	Take a few guesses before revealing the answer, including wrong guesses — the reasoning matters more than the score.
4	Reveal the flagged packet and connect it back to a pattern from today (unusual port, repeating connection, mismatched protocol).



Even a wrong guess forces participants to actually apply Source / Destination / Protocol / Port one more time before the day ends — that's the point, not getting it right.



10 End-of-Day Recap Questions

Q	Answer
Four fields that route a packet?	Source IP, Destination IP, Port, Protocol
Why does video call use UDP?	Speed over guaranteed delivery
RDP default port + main risk?	3389; brute-force / ransomware entry point
Two reasons IoT widens attack surface?	Weak default credentials; rarely patched
IDS vs IPS?	IDS detects & alerts; IPS detects & blocks
Firewall blocks a connection — infected?	No — blocked means stopped, not succeeded
Pattern suggesting a port scan?	Many attempts, one source, many ports/hosts, short time
Pattern suggesting a C2 beacon?	Regular repeating connections to same external address
Why watch DNS closely?	Spoofing, tunneling, amplification; reveals target domains
Key habit from Day 3?	Read the traffic before you trust it



11 Facilitator Preparation Checklist



Clean, safe sample PCAP for the walkthrough + Lab 1.



Separate PCAP for Lab 2 with a recognizable port scan + C2 beacon.



Build/test the HTML Packet Inspection Challenge (Lab 3).



Test Wireshark + PCAP loading on lab machines 24–48h before.



Check TryHackMe room access; pre-create accounts if needed.



Prepare individual checkpoint/exit-ticket questions per section.



Prepare the Firewall/IDS/IPS before/after animation assets.



Facilitator notes with expected answers + common misconceptions (e.g., “IDS blocks traffic”).



Confirm “Friday Request — Part 2” lines up with how Day 2 was actually delivered.



12 Final Recommended Day 3 Structure



One continuous thread: follow a packet → learn what it's made of → the rules it travels under → the protocols/devices generating it → the controls watching it → prove it in three solo labs → tie back into Day 2.

Approved Content Blocks



Opening hook — “Follow the Packet”



Network Fundamentals: OSI/TCP-IP, packets, IPs, ports



TCP vs UDP



Common Protocols: HTTP/HTTPS, DNS, SSH, SMB, RDP



IoT & Connected Devices (new)



Firewalls, IDS vs IPS



Wireshark intro



Lab 1, 2, 3 (individual)



TryHackMe companion practice



Final Investigation: Friday Request — Part 2



Closing Challenge: Spot the Suspicious Packet



Recap quiz + takeaways



R Planning References & Tool Notes

Reference	URL	Why it matters
Wireshark	wireshark.org	Primary tool for Lab 1 and Lab 2
TryHackMe – Network Fundamentals	tryhackme.com/module/network-fundamentals	Companion practice room
Wireshark Sample Captures	wiki.wireshark.org/SampleCaptures	Safe PCAPs for Lab 1
reveal.js	revealjs.com	Same deck engine as Day 2



Tool features and sample-capture availability can change — re-check before the workshop and keep a local backup of every PCAP used.

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